Unit V: Plant Physiology (Functional Organisation)

Chapter 14

Respiration

Learning Objectives

The learner will be able to,

- *Recognize the stages of glucose breakdown and its redox system.*
- *Differentiate aerobic respiration from anaerobic respiration.*
- *Describe the conditions under which respiration occurs.*
- *Realize the role of mitochondria as power house of the cell.*
- *Understand, how ATP molecules are generated during respiration.*

Chapter Outline

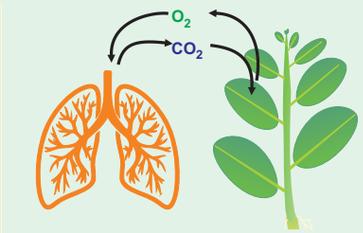
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14.1 Gaseous exchange
14.2 Structure of ATP
14.3 Redox reactions
14.4 Types of Respiration
14.5 Stages of Respiration
14.6 Respiratory Quotient
14.7 Anaerobic Respiration
14.8 Factors Affecting Respiration
14.9 Pentose Phosphate Pathway

If you are sleeping under a tree during night time you will feel difficulty in breathing. During night, plants take up oxygen and release carbon dioxide and as a result carbon dioxide will be abundant around the tree. This process of CO_2 evolution is called **respiration**. This process takes place during day time also (Figure 14.1). It is accompanied by breakdown of substrates and release of energy. In this chapter, respiration process in plants at cellular level will be dealt with.

Plant and Animal Interdependence

In biosphere, plants and animals are complementary systems which are integrated to sustain life. In plants, oxygen enters through the stomata and it is transported to cells, where oxygen is utilized for energy production. Plants require carbon dioxide to survive, to produce carbohydrates and to release oxygen through photosynthesis. These oxygen molecules are inhaled by human through the nose, which reaches the lungs where oxygen is transported through the blood and it reaches cells. Cellular respiration takes place inside the cell. A specialized respiratory system is present in animals but is absent in plants for delivering oxygen inside the cell. But the cellular respiration stages are similar in both plants and animals which hint at evolutionary divergence.



14.1 Gaseous Exchange

14.1.1 Respiration

The term respiration was coined by **Pepys** (1966). Respiration is a biological process in which oxidation of various food substances like carbohydrates, proteins and fats take place and as a result of this, energy is produced where O_2 is taken in and CO_2 is liberated. The organic substances which are oxidised during respiration are called respiratory substrates. Among these, glucose is the commonest respiratory substrate. Breaking of C-C bonds

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